quiz 3

set A

1) a) $(smoke \Rightarrow Fire) \Rightarrow (\neg smoke \Rightarrow \neg Fire)$

smoke	Fire	$(smoke \Rightarrow Fire)$	$\neg smoke$	$\neg Fire$	$(\neg smoke \Rightarrow \neg Fire)$	$(smoke \Rightarrow Fire) \Rightarrow (\neg smoke \Rightarrow \neg Fire)$
F	F	T	T	T	T	T
F	T	T	T	F	F	F
T	F	F	F	T	T	T
T	T	T	F	F	T	T

so The sentence is neither valid nor unsatisfiable. The sentence is satisfiable. Because it is true in some model.

(b) $Big \vee Dumb \vee (Big \Rightarrow Dumb)$

Big	Dumb	$Big \Rightarrow Dumb$	$Big \vee Dumb \vee (Big \Rightarrow Dumb)$
F	F	T	T
F	T	T	T
T	F	F	T
T	T	T	T

so the sentence is valid as it is true for all possible model.

2] a) $(A \Leftrightarrow B) \models (A \vee B)$ is true or correct

If $(A \vee B)$ is true the $(A \Leftrightarrow B)$ is also true ~~for every model of KB. we can~~ check it using truth table

A	B	~~$A \not\Leftrightarrow B$~~	$A \Leftrightarrow B$	$A \vee B$	
F	F	~~F~~	T	F	X
T	F	~~T~~	F	T	
F	T	~~T~~	F	T	
T	T	~~N~~	T	T	✓

~~here AVB is not true in every model so~~.

so, $A \Leftrightarrow B \models A \vee B$ is not correct.

b) $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable

a sentence is satisfiable if it is true in some model. from truth table we can see it.

A	B	$A \vee B$	$A \Rightarrow B$	$\neg(A \Rightarrow B)$	$(A \vee B) \wedge \neg(A \Rightarrow B)$
F	F	F	T	F	F
F	T	T	T	F	F
T	F	T	F	T	T
T	T	T	T	F	F

so the sentence is correct.

3]

$S_1 : \neg A$

$S_2 : \neg B \lor \neg C \lor A$

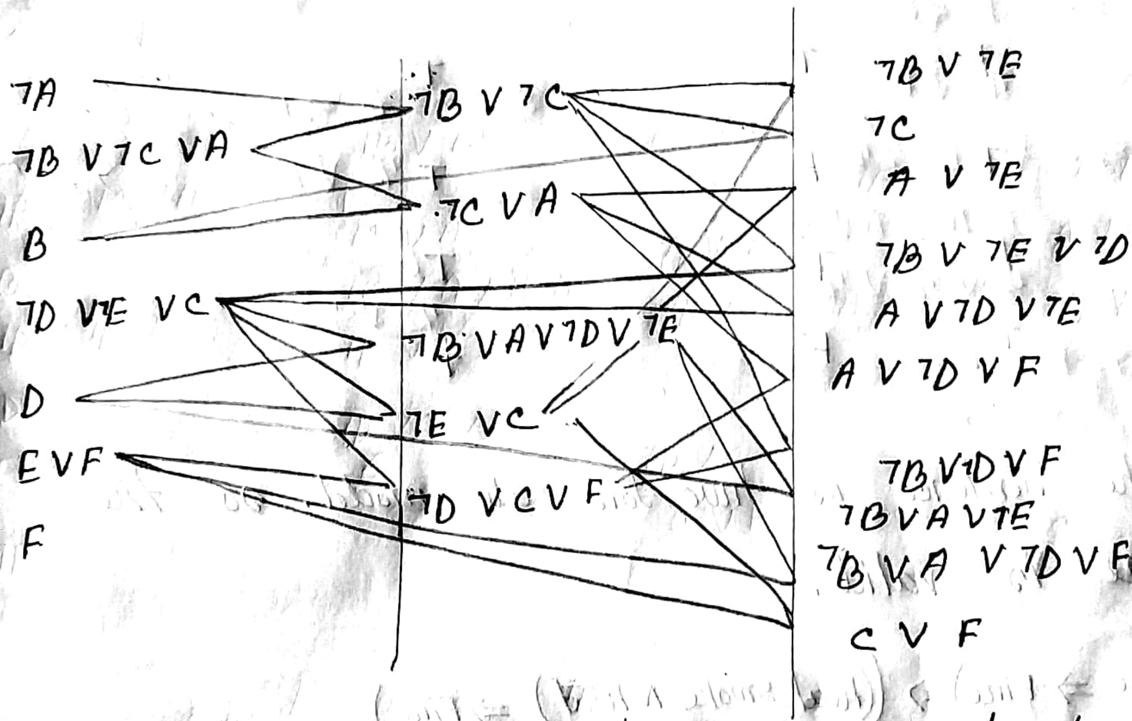
$S_3 : B$

$S_4 : \neg D \lor \neg E \lor C$

$S_5 : D$

$S_6 : E \lor F$

we have to prove $\neg F$. So we add F in set of clauses.



so ~~we can~~ hence we cannot find any empty clause (contradiction)

hence $\neg F$ can not entailed from KB.

Quiz 3
set B

1/ a) smoke $\vee$ Fire $\vee$ $\neg$Fire

smoke	Fire	$\neg$Fire	smoke $\vee$ Fire $\vee$ $\neg$Fire
F	F	T	T
F	T	F	T
T	F	T	T
T	T	F	T

as the sentence is true for all model. so the sentence is valid.

b) (smoke $\Rightarrow$ Fire) $\Rightarrow$ ((~~heat~~ smoke $\wedge$ heat) $\Rightarrow$ Fire)

smoke	Fire	heat	(smoke $\Rightarrow$ Fire)	(smoke $\wedge$ heat)	(smoke$\wedge$heat) $\Rightarrow$ Fire	
F	F	F	T	F	T ~~F~~	T
F	F	T	T	F	T	T
F	T	F	T	F	T	T
F	T	T	T	F	~~F~~T	T
T	F	F	F	F	T	T

$(smoke \Rightarrow Fire) \Rightarrow ((smoke \wedge heat) \Rightarrow Fire)$

smoke	fire	heat	$(smoke \Rightarrow Fire)$	$(smoke \wedge heat)$	$(smoke \wedge heat) \Rightarrow Fire$	$(smoke \Rightarrow Fire) \Rightarrow ((smoke \wedge heat) \Rightarrow Fire)$
T	F	T	F	T	F	F
T	T	T	T	T	T	T

as the sentence is true for some model, the sentence is satis-fiable.

2] a) $(A \wedge B) \models (A \Leftrightarrow B)$

A	B	$(A \wedge B)$	$A \Leftrightarrow B$
F	F	F	T
F	T	F	F
T	F	F	F
T	T	T	T ✓

so the sentence is correct as we know if $\alpha \models \beta$ then if α is true, β is true.

b) $(A \Leftrightarrow B) \models \neg A \vee B$

A	B	$\neg A$	$(A \Leftrightarrow B)$	$(\neg A \vee B)$
F	F	T	T	T ✓
F	T	T	F	T
T	F	F	F	F
T	T	F	T	F x

sentence is not correct because in $\alpha \models \beta$ if α is true then β is true

3)

S_1 : $A \Leftrightarrow (B \vee E)$

$(A \Rightarrow (B \vee E)) \wedge ((B \vee E) \Rightarrow A)$

$(\neg A \vee (B \vee E)) \wedge (\neg (B \vee E) \vee A)$

$(\neg A \vee B \vee E) \wedge ((\neg B \wedge \neg E) \vee A)$

$(\neg A \vee D \vee E) \wedge (\neg B \vee A) \wedge (\neg E \vee A)$

S_2 : $E \Rightarrow D$

$\neg E \vee D$

S_3 : $(C \wedge F) \Rightarrow \neg B$

$\neg (C \wedge F) \vee \neg B$

$\neg C \vee \neg F \vee \neg B$

S_4 : $E \Rightarrow B$

$\neg E \vee B$

$\cancel{S_5}$

S_5 : $B \Rightarrow F$

$\neg B \vee F$

S_6 : $B \Rightarrow C$

$\neg B \vee C$

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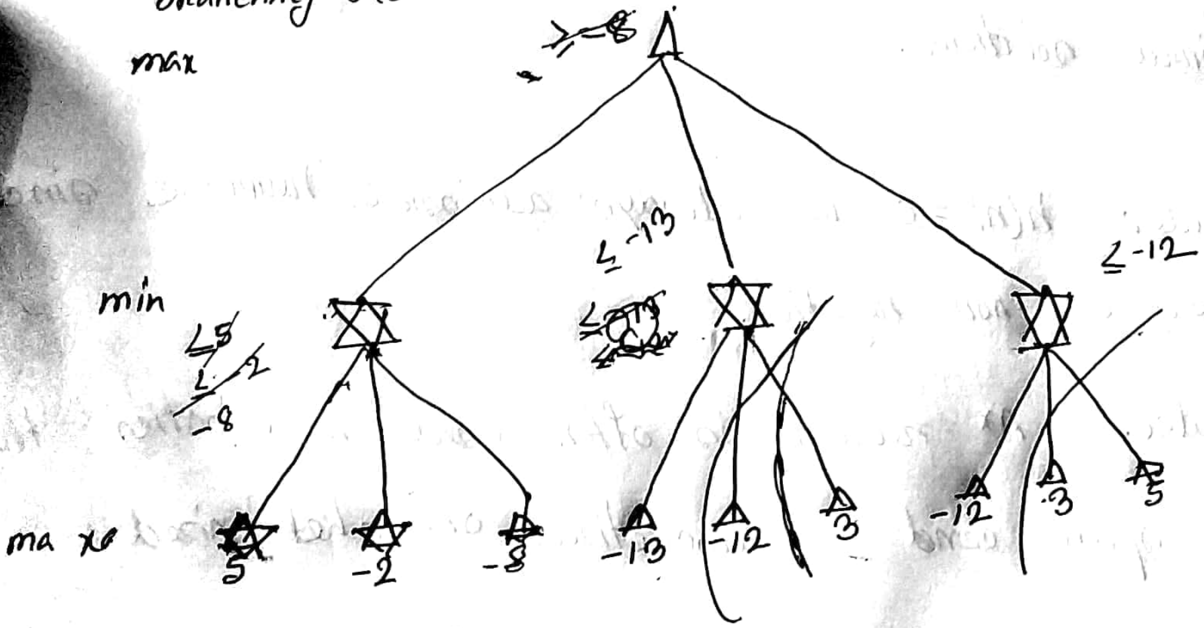
Quiz 2

set: A   sec: A

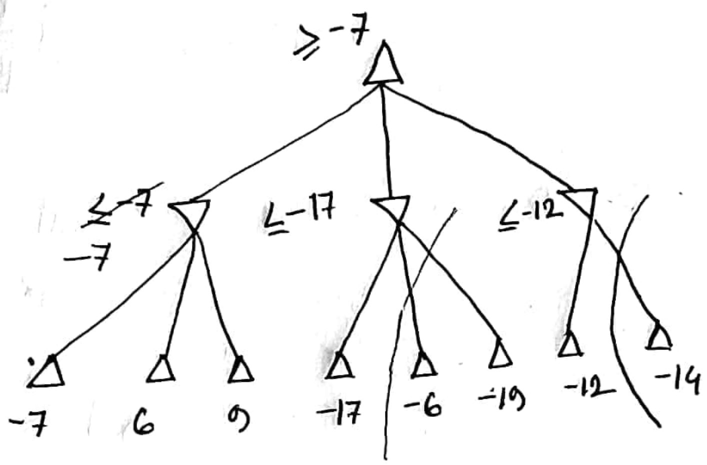
2| 5 -2 -8 -13 -12 3 -12 3 5

branching factor 3



Quiz 2
set B (section A)

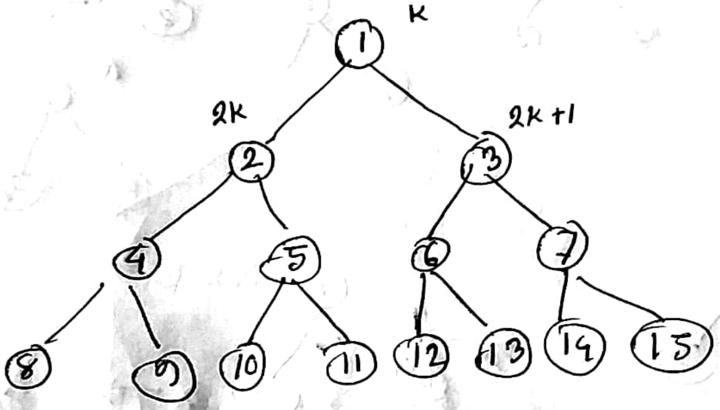
2) -7 6 9 -17 -6 -19 -12 -14



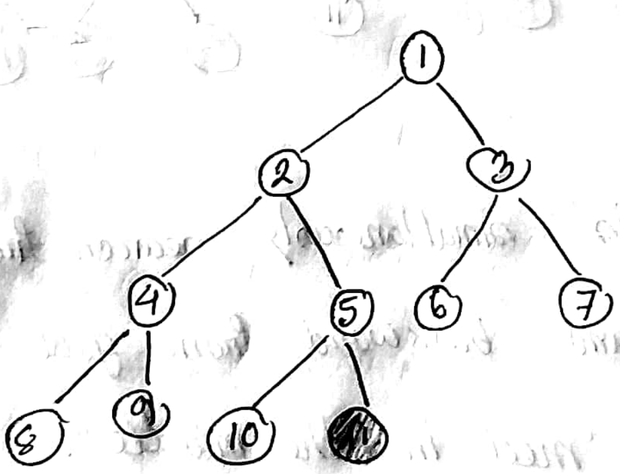
Quiz 1
set B
Sec A

9] ~~Patk Pattern databases store the exact solution for every~~

a.

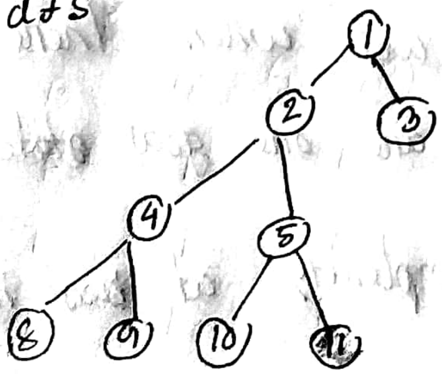


b) for bfs. goal 11



{~~1~~. ~~2~~, ~~3~~, ~~4~~, 5, 6, 7, 8, 9, 10, 11

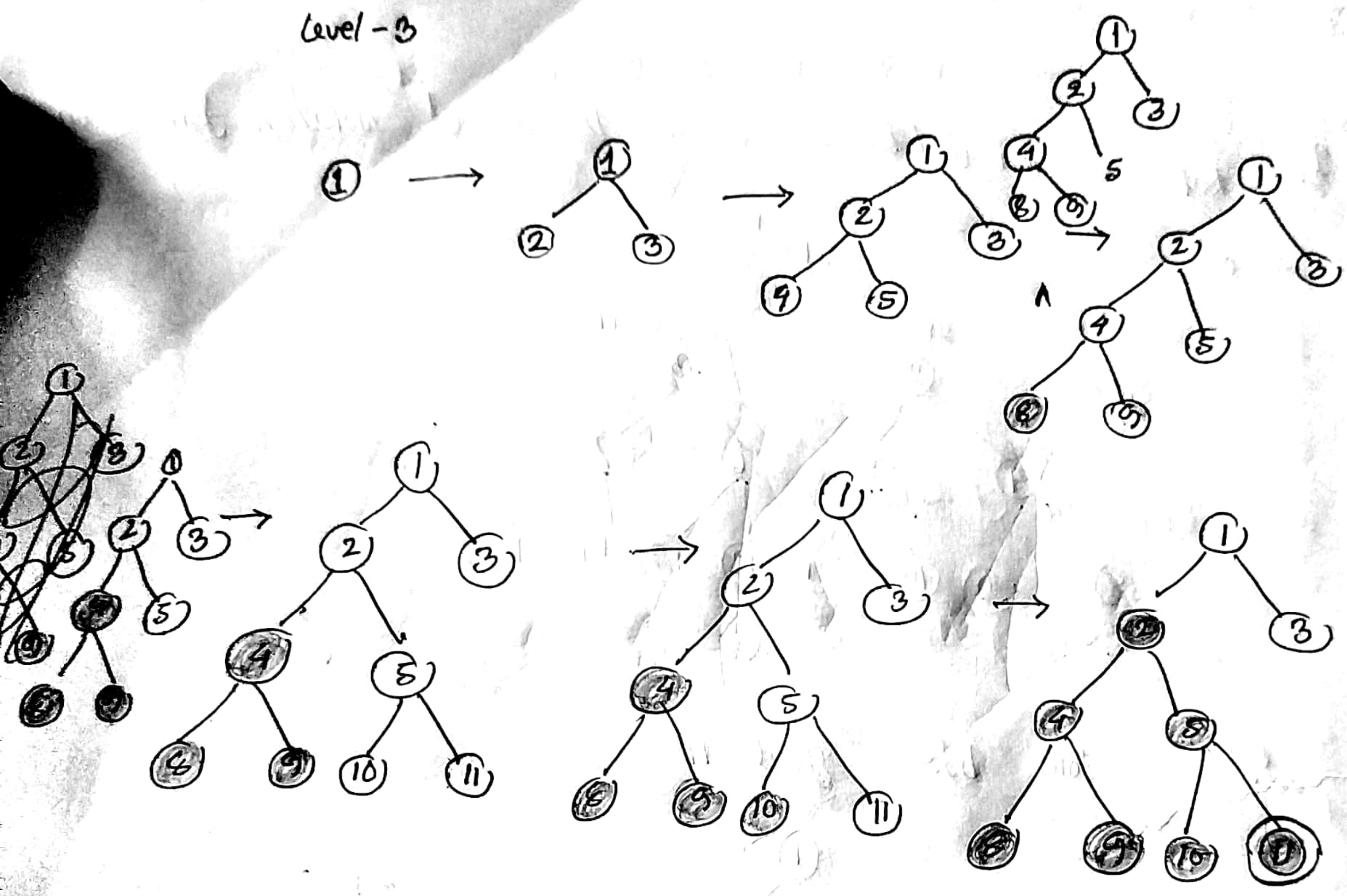
for dfs



{~~2~~, 3, ~~1~~
10, 11 ~~8~~, ~~9~~ ~~4~~, 5, 3

for iterative deepening search:

Level-3



d) Biderectional search is simultaneously search ~~from~~ forward from start node and backword from goal node. It stops when both "meet in the middle".

~~Branching factor~~ Bidirectional search performs well when there are single ~~simple~~ goal state. As there are one goal state so it would do better. Time complexity at best is $O(b^{d/2})$

where b is the branching factor. d is the depth.

1| a) ~~true~~ False. A lucky dfs might expand exactly d nodes to reach the goal. A* largely dominates any graph-search algorithm that is guaranteed to find optimal solutions.

b) True. $h(n) = 0$ is always admissible heuristic since cost are non negative.

c) False. A* search is often used in robotics; the space can be discretized or skeletonized.

d) True. Because BFS search is depends on the depth of the solution, not costs.

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